

Where star formation remains along the MAUVE RPS sequence

Audit and interpretation of SF / ND / NSF fractions, radial profiles, mass profiles, and spatial maps

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Primary notebook: 20260904_check_combined_SF_fraction_categories_by_stage_SNR_postfit.ipynb

Cross-check notebooks: the pre-peak, close-to-peak, and post-peak

20260901_check_*_SF_fraction_and_rSFMS_categories.ipynb notebooks

Question addressed here: how the *spatial occupancy* of the SF, non-detected (ND), and detected non-SF (NSF) classes changes from pre-peak to close-to-peak to post-peak. The intensity of the surviving SF regions is deliberately deferred.

Executive answer

The present data support a strong first-order result, but not the simplest one-line version of it.

The resolved SF footprint contracts along the RPS sequence, and the strongest secure pre-to-post change is a galaxy-level transfer from SF-classified area to Balmer non-detection. The transformation is spatially differentiated: outer / low- Σ_* regions preferentially become ND, whereas central high- Σ_* regions remain emission-line detected but are increasingly NSF.

The geometry is therefore **not pure uniform quenching**, but it is also not adequately summarized by one increasingly negative radial slope. The more faithful description is:

1. **Pre-peak to close-to-peak:** onset of outside-in erosion. The decline in f_{SF} is larger at $1-2R_e$ than inside R_e , and the outer loss is accompanied mainly by increasing f_{ND} .
2. **Close-to-peak to post-peak:** a large additional loss of SF occupancy across the observed disc. Inside $2R_e$ its additive size is nearly uniform, but its observational destination remains radius dependent: NSF is important centrally and ND dominates outside.
3. **Overall pre-peak to post-peak:** a contracting SF edge plus a large normalization decrease. An illustrative half-occupancy radius contracts from about $2.16R_e$ to $1.63R_e$ to $0.95R_e$ after the post-fit S/N cut.

At the individual-galaxy level, the **pre/post separation is exceptionally consistent** in the retained footprint: all eight pre-peak systems have higher whole-footprint f_{SF} than all thirteen post-peak systems, and all have lower f_{ND} . The close-to-peak group does not form a narrow intermediate sequence; it contains both pre-like and post-like galaxies. The radial shapes are also heterogeneous: some post-peak galaxies show dramatic truncation, while others are already SF-poor everywhere or retain rings/asymmetric patches. Therefore the stage curve is a population summary, not a universal evolutionary track followed by every galaxy.

The NSF fraction does **not** show a secure monotonic stage trend. Its spatial redistribution is interesting, but its physical interpretation requires BPT/EW/dispersion and morphological separation. NGC4383 remains a major pre-peak NSF outlier even after the S/N cut.

The $S/N_{\text{postfit}} \geq 25$ selection does remove the faint NGC4383 footprint, but it is not a neutral or NGC4383-specific cleanup. It removes 77.3% of NGC4383 and preferentially removes ND pixels across the full sample. The post-cut curves are conditional on the retained footprint and should be presented as a sensitivity analysis alongside the uncut result, not as the sole occupancy measurement.

1. What the linked discussion is really asking

The linked meeting discussion frames Paper 2 Q1 as a decomposition of environmental transformation into three observables:

$$\begin{aligned} \text{resolved transformation} = & \text{change in SF occupancy} \\ & + \text{change in surviving-SF intensity} \\ & + \text{change in ionisation state.} \end{aligned}$$

The current task is the first term: **where SF remains**. The fraction plots measure whether usable spaxels change observational class. They do not measure whether a spaxel that remains SF has become weaker or stronger in Σ_{SFR} .

The meeting's key distinction between normalization and shape is correct:

- A near-constant downward offset in $f_{\text{SF}}(R)$ resembles a disc-wide loss of SF occupancy.
- A difference that becomes progressively more negative with radius resembles outside-in transformation.
- A difference strongest at small radius resembles inside-out transformation.

However, bounded fraction profiles that reach zero do not remain linear. In a truncated disc, the pre- and post-peak curves can converge again at large radius because both have little SF. Consequently, a full-range straight-line slope can become *shallower* after severe truncation. A radius at which the curve turns down or crosses a fixed relative level is often a better shape diagnostic.

The meeting also explicitly requires four checks before interpreting a stacked curve:

1. show how many galaxies support each bin;
2. inspect galaxies one by one;
3. repeat the result without NGC4383;
4. separate the occupancy result from the later surviving-SF intensity analysis.

Those requirements guide this report.

2. Evidence and definitions

2.1 Sources inspected

- Linked ChatGPT meeting synthesis, including the underlying meeting transcript available in the conversation.
- `20260904_check_combined_SF_fraction_categories_by_stage_SNR_postfit.ipynb`, including code, stored tables, all 52 stage-plus-individual curve figures, and all 26 stored spatial maps.
- `20260901_check_pre_peak_SF_fraction_and_rSFMS_categories.ipynb`.
- `20260901_check_close_to_peak_SF_fraction_and_rSFMS_categories.ipynb`.
- `20260901_check_post_peak_SF_fraction_and_rSFMS_categories.ipynb`.
- The current live FITS products used by the combined notebook were read through a calculation-only execution of its data cells; none of the four source notebooks was modified.

The combined notebook has twelve sequentially executed code cells and no stored error outputs. Its bookkeeping checks end with `FINAL_VERIFICATION_OK`. That verifies category accounting and weights; it does not test the statistical significance or physical geometry of the stage differences.

2.2 Category partition

For each galaxy, the valid-disc footprint is

$$V = \{\text{finite } \Sigma_{\star}\} \cap \{\text{finite gas BINID}\} \cap \{\text{MASK} = 0\} \cap \{\text{finite } R/R_e\}.$$

The unchanged SF selection is

$$S = V \cap \{\text{finite } \Sigma_{\text{SFR, HII}}\} \cap \{\text{EW}(\text{H}\alpha) > 6 \text{ \AA}\} \cap \{\sigma_{\text{H}\alpha, \text{int}} < 45 \text{ km s}^{-1}\}.$$

Each Balmer line is detected when its flux and positive error are finite, $F/\sigma_F \geq 3$, and $F \geq 20$ in the pipeline map units. With $D_B = D_{\text{H}\alpha} \cap D_{\text{H}\beta}$,

$$\begin{aligned} \text{SF} &= S, \\ \text{ND} &= V \cap \neg D_B, \\ \text{NSF} &= V \cap D_B \cap \neg S. \end{aligned}$$

The code asserts pixel by pixel that

$$f_{\text{SF}} + f_{\text{ND}} + f_{\text{NSF}} = 1$$

over the chosen denominator.

Interpretive limits are essential:

- **ND is a sensitivity-defined Balmer outcome**, not proof that SFR is exactly zero. A large fraction of ND pixels are H α -detected but H β -failed.
- **NSF is detected but fails the HII/EW/dispersion selection**. It can include DIG, shocks, outflow emission, LI(N)ER/AGN-like excitation, old-star ionisation, or mixtures. It is not a single physical evolutionary state.

2.3 Sample

The live primary sample requires inclination $< 80^\circ$ and complete products. It contains 26 product-level systems:

Stage	Passing / candidates	Systems used
Pre-peak	8 / 8	NGC4189, NGC4254, NGC4294, NGC4321, NGC4351, NGC4383, NGC4535, NGC4567_8
Close-to-peak	5 / 5	NGC4298, NGC4424, NGC4501, NGC4654, NGC4694
Post-peak	13 / 16	IC3392, NGC4064, NGC4293, NGC4380, NGC4394, NGC4405, NGC4419, NGC4450, NGC4457, NGC4569, NGC4580, NGC4606, NGC4698

NGC4548, NGC4579, and NGC4689 are excluded because the required mass, SFR, EW, mask, and SFH products are absent. `NGC4567_8` is one product representing two physical components; its radius is the minimum component R/R_e .

2.4 Estimators

The report treats galaxies as the independent environmental units. The primary stage estimator is therefore

$$f_{c,b}^{\text{equal}} = \frac{1}{G_b} \sum_{g \in b} \frac{N_{c,g,b}}{N_{\text{valid},g,b}},$$

where a galaxy enters bin b only if it has at least 50 valid pixels there. The pooled-spaxel estimator

$$f_{c,b}^{\text{pooled}} = \frac{\sum_g N_{c,g,b}}{\sum_g N_{\text{valid},g,b}}$$

is retained as a description of all sampled spaxels, but it is not the preferred population-level environmental comparison.

The combined notebook uses common 0.25-dex $\log\Sigma_\star$ bins and 0.25- R_e radial bins. For the derived figures in this report, a displayed stage bin additionally needs at least 1,000 retained pixels and at least three contributing galaxies. Whole-footprint uncertainty intervals use 20,000 percentile bootstrap resamples of complete galaxies within each stage. These resamples were added for this report; they are not present in the source notebook.

3. The global stage change: SF is primarily transferred to ND

3.1 Full valid footprint, before the post-fit S/N cut

Estimator	Stage	f_{SF}	f_{ND}	f_{NSF}
Equal galaxy	Pre	0.388	0.312	0.300
Equal galaxy	Close	0.251	0.509	0.239
Equal galaxy	Post	0.066	0.727	0.206
Pooled spaxels	Pre	0.458	0.244	0.297
Pooled spaxels	Close	0.360	0.402	0.239
Pooled spaxels	Post	0.063	0.714	0.223

Both estimators give the same qualitative ordering: f_{SF} falls and f_{ND} rises strongly from pre to post. The pooled close-to-peak f_{SF} is higher because the large, blue-rich NGC4501 and NGC4654 footprints contribute many more pixels. This is exactly why equal-galaxy inference is needed.

3.2 Conditional on $S/N_{\text{postfit}} \geq 25$

Stage	f_{SF}	95% galaxy-bootstrap interval	f_{ND}	95% interval	f_{NSF}	95% interval
Pre	0.494	0.396--0.596	0.137	0.050--0.237	0.369	0.275--0.487
Close	0.332	0.107--0.556	0.386	0.199--0.578	0.282	0.215--0.354
Post	0.101	0.054--0.152	0.628	0.572--0.689	0.272	0.212--0.336

The whole-galaxy difference intervals are:

Transition (later minus earlier)	Δf_{SF}	Δf_{ND}	Δf_{NSF}
Close - pre	-0.162 [-0.404, +0.079]	+0.249 [+0.039, +0.472]	-0.087 [-0.219, +0.032]
Post - close	-0.231 [-0.456, -0.006]	+0.242 [+0.030, +0.440]	-0.011 [-0.106, +0.082]
Post - pre	-0.393 [-0.506, -0.284]	+0.491 [+0.378, +0.596]	-0.098 [-0.228, +0.017]

The secure global signal is therefore the SF-ND exchange, especially pre to post. NSF is consistent with no monotonic change. These are cross-sectional stage comparisons, not measurements of the same galaxies at three times.

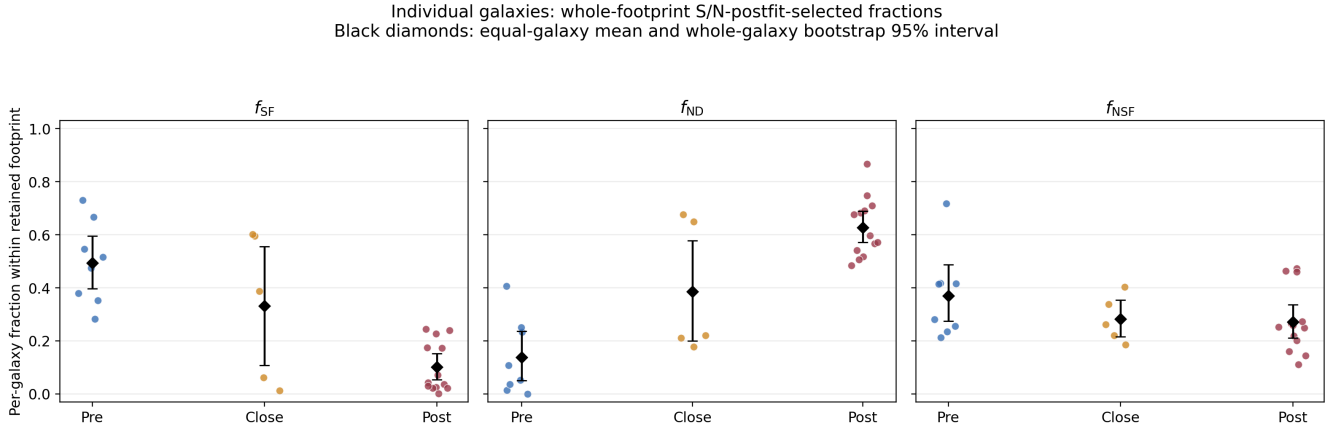


Figure 1. Per-galaxy global fractions after the post-fit S/N cut. Black diamonds and bars show equal-galaxy means and whole-galaxy bootstrap intervals added for this report.

4. The S/N cut is useful, but it changes the question

The cut retains 70.9% of pre-peak, 78.7% of close-to-peak, and 76.3% of post-peak originally valid pixels. Its action is strongly category dependent:

Stage	SF retained	ND retained	NSF retained
Pre	0.813	0.379	0.819
Close	0.960	0.517	0.980
Post	0.958	0.676	0.986

It preferentially removes ND rather than removing each class at the same rate. Relative to the uncut equal-galaxy means, it changes the conditional fractions by:

Stage	Δf_{SF}	Δf_{ND}	Δf_{NSF}
Pre	+0.106	-0.175	+0.069
Close	+0.081	-0.123	+0.042
Post	+0.034	-0.100	+0.065

This is expected if low-continuum-S/N outskirts contain many weak or absent Balmer detections. It also means that the cut can censor the very regions used to diagnose outside-in erosion.

The S/N variable comes from the stellar-population post-fit product, not from the Balmer-line detection rule. The cut is defensible when reliable $\Sigma_\star$ placement is required, but its scientific denominator must be stated explicitly:

$$f_c^{\text{cut}} = \frac{N_c(S/N_{\text{postfit}} \geq 25)}{N_{\text{valid}}(S/N_{\text{postfit}} \geq 25)}.$$

It is *not* the fraction of the original valid disc in category c . The maps preserve the difference by painting excluded pixels gray; the curves condition them away.

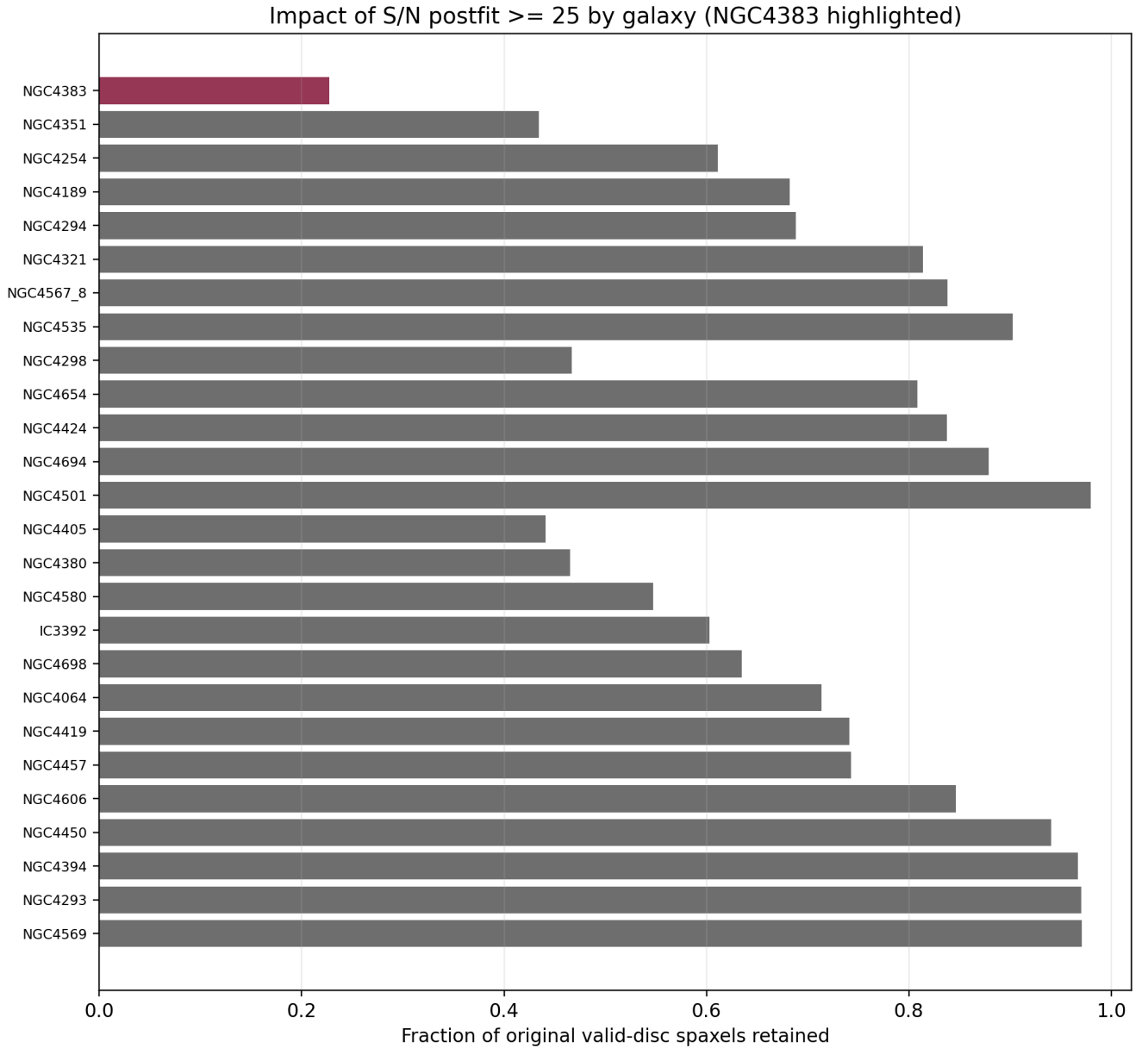


Figure 2. Fraction of each galaxy's original valid-disc footprint retained. NGC4383 is the strongest case, but several other galaxies also lose roughly half their footprint.

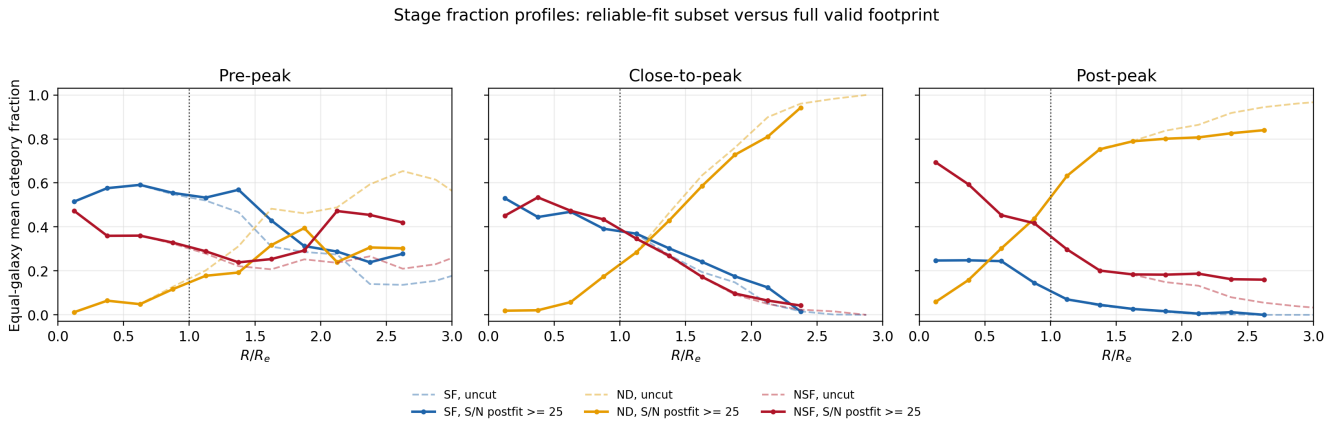
A clean final presentation should therefore show either:

- uncut and cut profiles together; or

- four fractions relative to the original denominator: SF, ND, NSF, and low-S/N excluded.

The stage profiles below explicitly compare the two denominators.

5. Radial result: contracting edge plus later disc-wide loss



5.1 Shape and normalization

The three f_{SF} profiles are not parallel translations:

- **Pre-peak:** $f_{\text{SF}} \simeq 0.52\text{--}0.59$ to about $1.4R_e$, then declines.
- **Close-to-peak:** the central value remains about 0.53, but the curve declines steadily and reaches near zero by about $2.2R_e$.
- **Post-peak:** the central plateau is only about 0.25 and the curve falls below 0.1 near $1.0R_e$.

An illustrative edge statistic makes this clearer. Define the inner baseline as the mean of the first three radial bins ($R/R_e = 0.125, 0.375, 0.625$) and take the first outward crossing of half that value after the profile maximum:

Stage	Inner baseline f_{SF}	Half-baseline radius, cut	Half-baseline radius, uncut
Pre	0.561	$2.16R_e$	$2.00R_e$
Close	0.481	$1.63R_e$	$1.48R_e$
Post	0.247	$0.95R_e$	$0.95R_e$

These are descriptive crossings, not fitted physical truncation radii, but their monotonic contraction is the clearest current evidence for outside-in evolution.

A single straight slope gives the misleading sequence -0.151 , -0.215 , and -0.129 per R_e for pre, close, and post over the common $0.125\text{--}2.375 R_e$ bins. The post slope becomes shallower because f_{SF} has already hit its floor. Thus the meeting's request to examine “slope” should be implemented as **shape/edge analysis**, not necessarily a one-parameter linear regression.

5.2 Inner and outer annuli

Equal-galaxy fractions integrated within broad annuli are:

Stage	Region	f_{SF}	f_{ND}	f_{NSF}
Pre	$R < R_e$	0.567	0.077	0.356
Pre	$1 \leq R/R_e < 2$	0.507	0.200	0.293
Close	$R < R_e$	0.434	0.099	0.466
Close	$1 \leq R/R_e < 2$	0.294	0.469	0.238
Post	$R < R_e$	0.200	0.318	0.482
Post	$1 \leq R/R_e < 2$	0.058	0.729	0.213

These values reveal the transition more directly:

Transition	Inner Δf_{SF}	Outer Δf_{SF}	Inner Δf_{ND}	Outer Δf_{ND}	Inner Δf_{NSF}	Outer Δf_{NSF}
Close - pre	-0.133	-0.213	+0.023	+0.268	+0.110	-0.055
Post - close	-0.234	-0.235	+0.218	+0.260	+0.016	-0.025
Post - pre	-0.367	-0.449	+0.241	+0.528	+0.126	-0.080

The interpretation is:

- The **pre-to-close** change is preferentially outside-in. The outer SF loss is larger, and almost all of the corresponding outer gain is ND. The centre changes differently: NSF increases while ND remains low.
- The **close-to-post** SF loss is almost identical in these two broad annuli. It resembles an additional normalization decrease after the close sample has already developed an outside-in gradient. Nevertheless, the destination remains spatially differentiated: ND grows outside and NSF remains important inside.
- The **full pre-to-post** difference is stronger outside but substantial everywhere. “Outside-in plus a disc-wide reduction” is more accurate than either “pure outside-in” or “pure uniform.”

Where the category fractions change along the RPS sequence (S/N postfit ≥ 25)

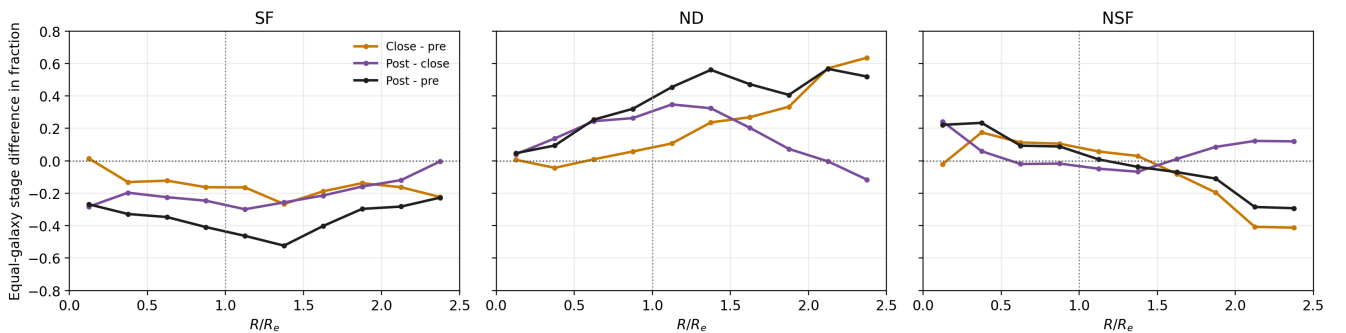


Figure 4. Later-minus-earlier equal-galaxy fraction profiles. The non-monotonic differences show why one full-range slope is insufficient. The strongest post-minus-pre SF deficit occurs near the transition region around $1\text{--}1.5R_e$, then diminishes where both curves approach the floor.

6. Stellar-surface-density result: low- $\Sigma_\star$ loss and high- $\Sigma_\star$ NSF

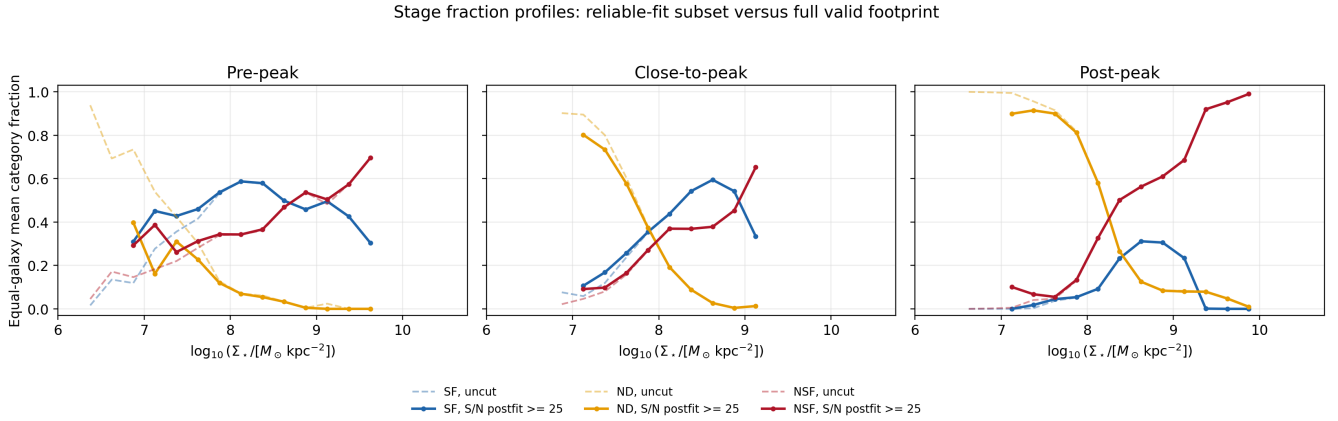


Figure 5. Equal-galaxy profiles versus stellar surface density, with cut and uncut denominators shown separately.

The low- $\Sigma_\star$ behaviour agrees with the radial result:

$\log \Sigma_\star$	Stage	f_{SF}	f_{ND}	f_{NSF}	Contributing galaxies
7.125	Pre	0.451	0.162	0.387	5
7.125	Close	0.107	0.802	0.091	5
7.125	Post	0.000	0.899	0.101	5
8.125	Pre	0.587	0.070	0.343	8
8.125	Close	0.438	0.192	0.370	5
8.125	Post	0.093	0.580	0.327	13
8.625	Pre	0.499	0.033	0.469	7
8.625	Close	0.595	0.027	0.378	5
8.625	Post	0.311	0.125	0.563	13
9.125	Pre	0.496	0.000	0.504	6
9.125	Close	0.334	0.013	0.653	4
9.125	Post	0.234	0.080	0.686	12

At low $\Sigma_\star$, the stage change is overwhelmingly SF to ND. At high $\Sigma_\star$, ND becomes rare and NSF becomes the dominant complement to SF. This is consistent with weakly bound outer discs being lost below line-detection thresholds while central emission remains detectable but often fails the HII-like selection.

The mass profile is not an independent proof of radial outside-in transformation. $\Sigma_\star$ and radius covary, the contributing-galaxy set changes across bins, and high- $\Sigma_\star$ points contain bulges/nuclei where NSF has multiple possible origins. The correct next test is a joint or matched comparison of stage, radius, and $\Sigma_\star$.

7. Spatial maps: the transition is not azimuthally uniform

The spatial maps add information that one-dimensional profiles erase. Gray shows S/N-excluded parts of the original valid footprint; blue, orange, and red are measured only in the retained footprint.

7.1 Pre-peak

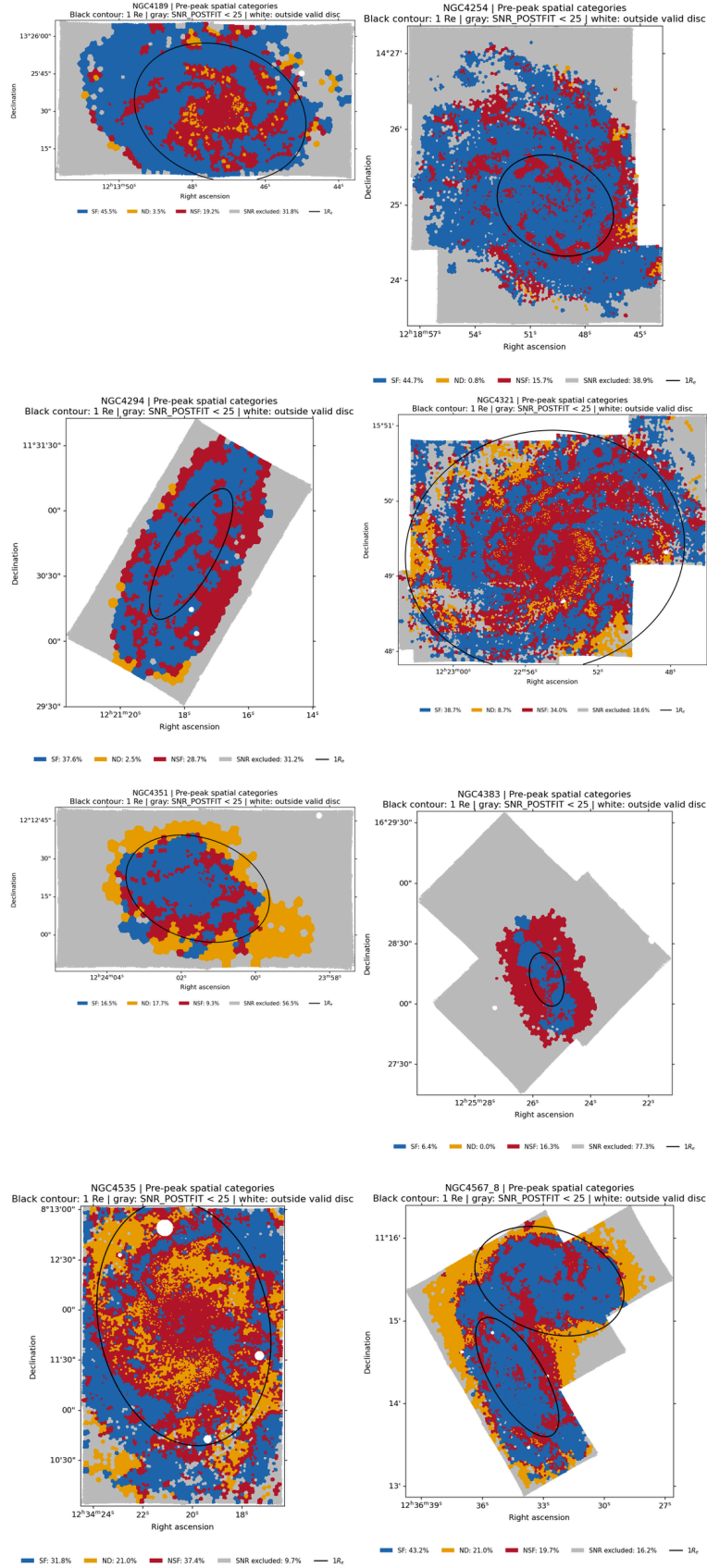


Figure 6. Reused stored maps from cell 26 of the combined notebook.

Most pre-peak systems retain blue SF over a large fraction of the disc. Peripheral orange is already visible in NGC4351 and NGC4567_8, and NGC4294 has a strong red outer component. NGC4535 is patchy. NGC4383 is qualitatively exceptional: its retained map is red dominated and most of its original footprint is gray.

7.2 Close-to-peak

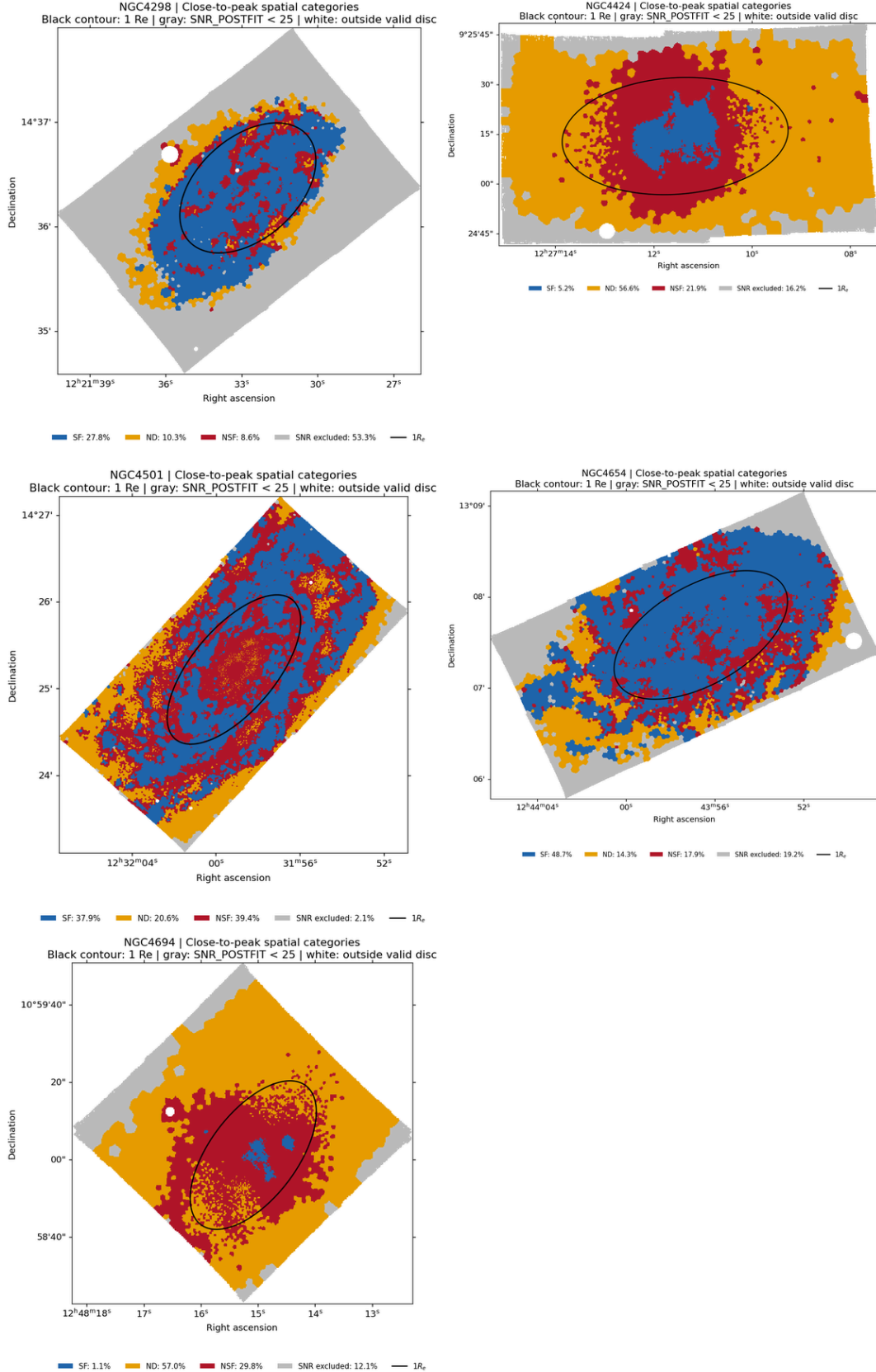


Figure 7. Reused stored close-to-peak maps.

The close stage visibly contains two regimes:

- NGC4298, NGC4501, and NGC4654 remain blue rich, with orange becoming more important toward edges or one side.
- NGC4424 and NGC4694 have compact central SF embedded in red and surrounded by extensive orange.

This is why the close-stage bootstrap interval is wide and why it should be described as a heterogeneous bridge, not a universal intermediate curve.

7.3 Post-peak

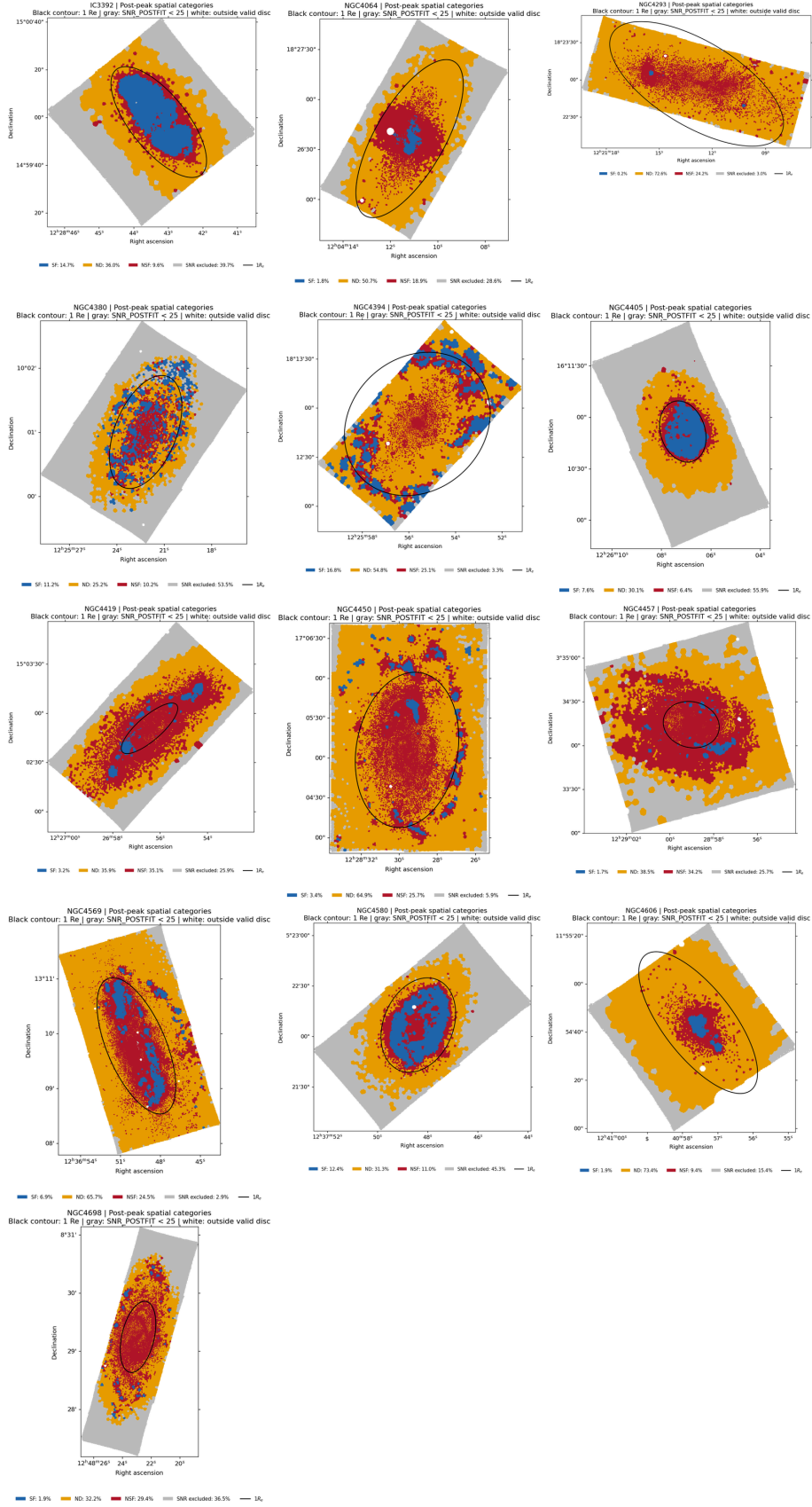


Figure 8. Reused stored post-peak maps.

The common post-peak pattern is a small central or patchy blue component, a large orange outer component, and red central/intermediate emission. Strongly truncated examples include IC3392, NGC4405, NGC4569, and NGC4580. NGC4293, NGC4064, NGC4606, and NGC4698 are nearly SF-free globally. NGC4394 and several other systems retain rings, edge patches, bars, or asymmetric structures that do not reduce to a simple radial law.

Therefore “uniform” in an azimuthally averaged annulus would not imply a spatially uniform physical process. One-sided stripping can produce a modest annular change if the opposite side remains blue, and a ring can give a positive outer-minus-inner contrast even in a strongly transformed galaxy.

8. Do individual galaxies follow the stage trend?

8.1 Global occupancy: yes for pre versus post, only partly for adjacent stages

For cut-footprint global fractions:

- $P(f_{\text{SF, pre}} > f_{\text{SF, post}}) = 1.000$ over all 104 pre/post galaxy pairs.
- $P(f_{\text{ND, pre}} < f_{\text{ND, post}}) = 1.000$.
- $P(f_{\text{SF, pre}} > f_{\text{SF, close}}) = 0.625$.
- $P(f_{\text{SF, close}} > f_{\text{SF, post}}) = 0.723$.

Thus the endpoints are not driven by one large galaxy. The adjacent-stage ordering is much less deterministic, especially because the close group spans from $f_{\text{SF}} = 0.013$ (NGC4694) to 0.602 (NGC4654).

NSF does not follow a monotonic sequence. Its between-galaxy scatter and spatial location are more informative than its global stage mean.

8.2 Radial shape: heterogeneous and affected by floor effects

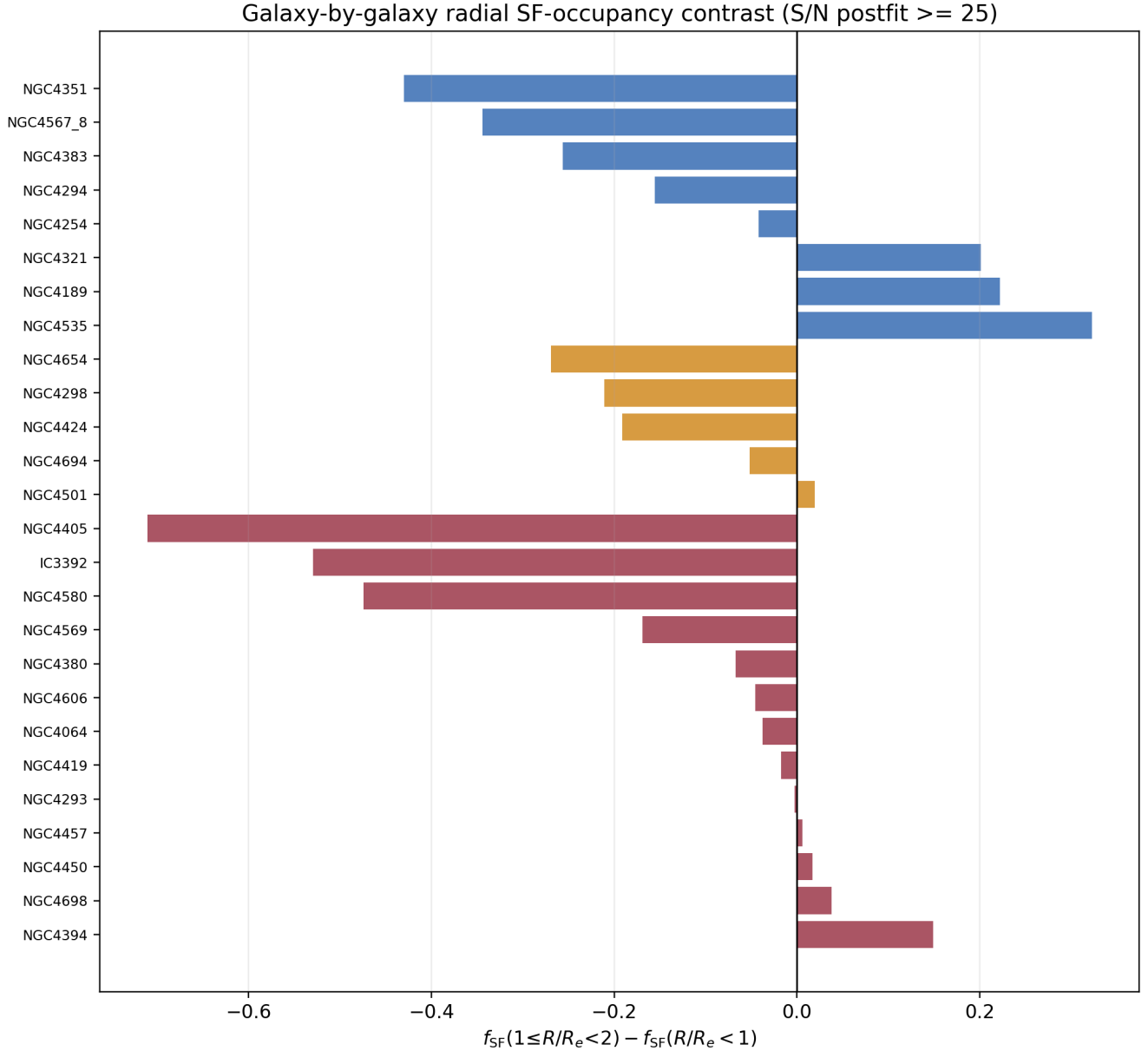


Figure 9. Per-galaxy difference between the outer ($1-2R_e$) and inner ($< R_e$) SF fractions. Negative values mean less SF occupancy outside.

Negative contrasts occur in 5/8 pre, 4/5 close, and 9/13 post systems. The means are -0.060 , -0.141 , and -0.142 , but the post median is only -0.038 . The post mean is pulled by a subset with very strong truncation: NGC4405 (-0.711), IC3392 (-0.530), NGC4580 (-0.474), and NGC4569 (-0.169).

Several post systems have near-zero contrast because they have very little SF in either annulus, not because their discs are untransformed. NGC4394 has a positive contrast because its residual SF is ring/edge weighted. This again shows why one radial slope or contrast cannot be the only stage statistic.

The defensible individual-galaxy conclusion is:

Galaxies broadly follow the stage trend in total SF occupancy, especially at the pre/post endpoints, but they do not follow one universal radial profile. Close-to-peak is a physical mixture, and post-peak combines strongly truncated discs, globally SF-poor floor cases, and residual ring/asymmetric morphologies.

9. NGC4383 and the intended S/N diagnostic

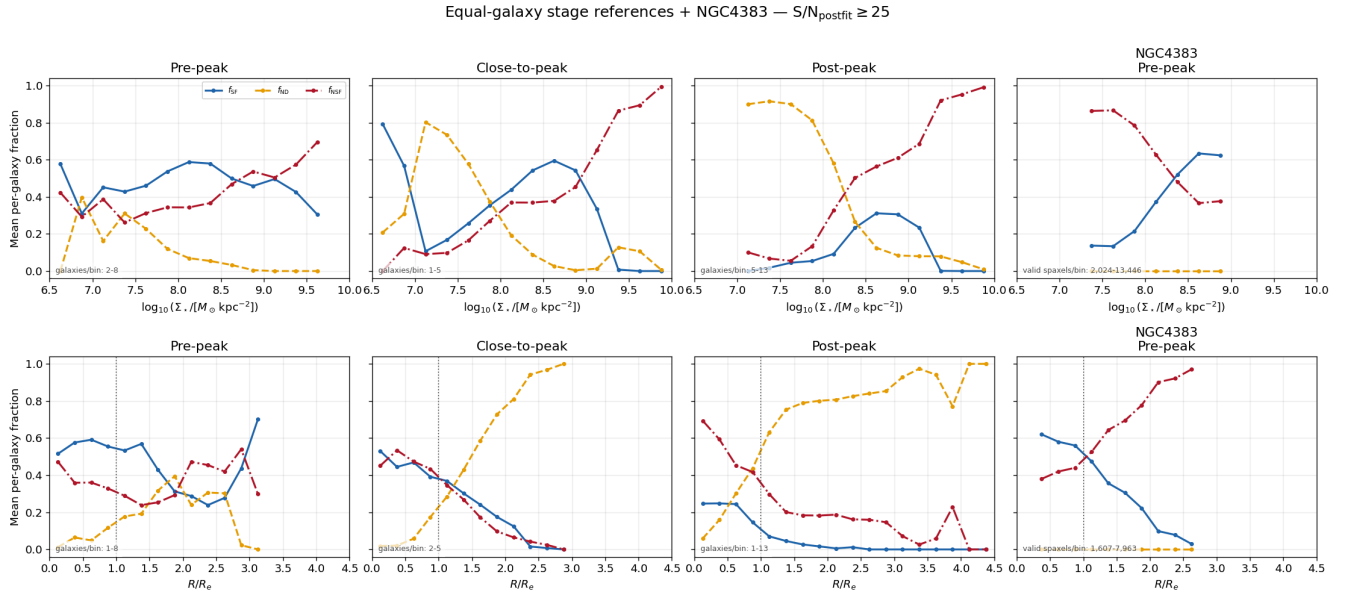


Figure 10. Reused equal-galaxy reference plus NGC4383 figure from cell 21.

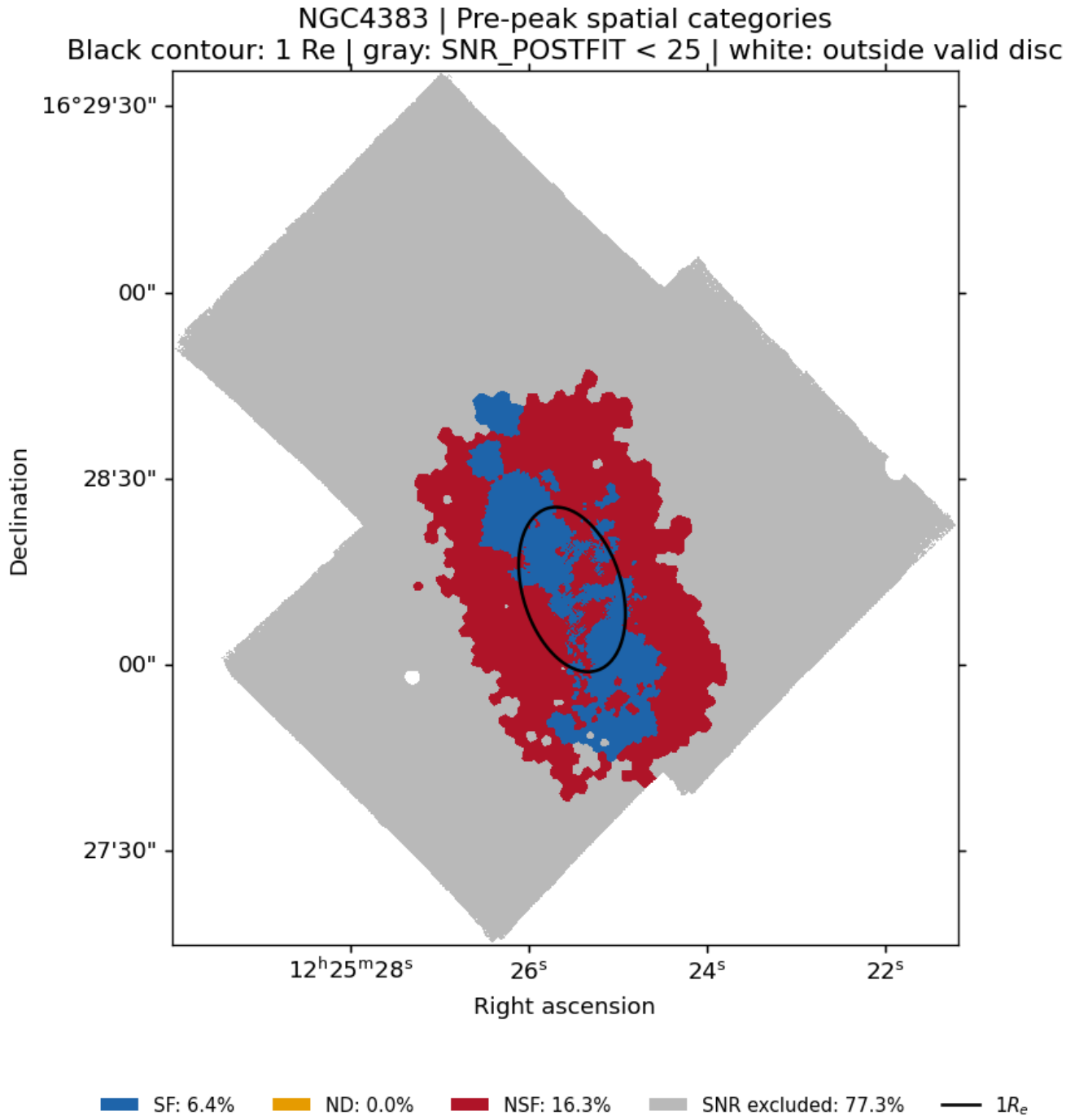


Figure 11. Reused NGC4383 map from cell 26.

The cut does what was intended in one important sense: it removes 175,482 of 227,031 originally valid NGC4383 pixels, leaving only 22.7%. The removed region includes most of the faint extended footprint.

It does **not** remove NGC4383 as an NSF influence. Within the retained denominator,

$$(f_{\text{SF}}, f_{\text{ND}}, f_{\text{NSF}}) = (0.282, 0.000, 0.718).$$

Relative to the original footprint, the map consists of 6.4% SF, 0.0% ND, 16.3% NSF, and 77.3% gray excluded area. The retained red component still surrounds/intersects the blue disc and is spatially compatible with the known exceptional outflow morphology, but the category map alone cannot prove its mechanism.

The explicit leave-NGC4383-out result is:

Denominator	Pre sample	f_{SF}	f_{ND}	f_{NSF}
Uncut	all 8	0.388	0.312	0.300
Uncut	without NGC4383	0.428	0.315	0.257
S/N cut	all 8	0.494	0.137	0.369
S/N cut	without NGC4383	0.524	0.157	0.319

Removing it changes individual radial NSF bins by as much as 0.156. The primary pre/post SF--ND conclusion survives, but detailed NSF shape claims do not pass a leave-one-out standard yet.

There is also a bookkeeping correction to the meeting synthesis: **NGC4383 is pre-peak**, so it cannot cause a feature in a post-only curve. It can affect the pre-peak reference, all-stage curves, and any later-minus-pre difference.

The current combined notebook applies the S/N mask but never performs the meeting's requested complete NGC4383 exclusion. That is a genuine missing analysis.

10. Which toy geometry best matches the data?

Toy scenario	Expected signature	Match to current data
Pure uniform occupancy loss	Nearly constant $\Delta f_{\text{SF}}(R)$; no edge contraction	Incomplete. Close-to-post has an approximately equal inner/outer decrement, but the SF edge still contracts strongly and the destination differs with radius.
Pure outside-in loss	Central f_{SF} preserved; increasingly negative difference outward; outer SF becomes ND	Core component, especially pre-to-close. Low- $\Sigma_{\star}$ /outer ND growth and the contracting half-occupancy radius support it. It is not the whole post-peak story.
Pure inside-out loss	Central deficit stronger than outer deficit	Not dominant globally. Central SF loss exists, but it mostly appears as detected NSF and coexists with strong outer ND. Some individual morphologies can look inside-out or ring-like.
Outside-in truncation plus disc-wide loss	Contracting edge plus lower normalization; radial destination differs	Best current description. Pre-to-close is preferentially outside-in; close-to-post adds a large disc-wide occupancy loss.
Outside-in stripping plus central/other ionisation	Outer SF $\rightarrow$ ND and central SF $\rightarrow$ NSF	Consistent with the maps and annuli, but mechanism unproven. NSF must be decomposed before assigning shocks, AGN, old stars, or outflow physics.

The safest wording for a paper or meeting is:

The MAUVE stage sequence is consistent with progressive contraction of the SF-classified footprint. The onset from pre-peak to close-to-peak is preferentially outside-in, while the post-peak population shows an additional large loss of SF occupancy throughout the inner two effective radii. The observational endpoint depends on location: Balmer non-detections dominate at low stellar surface density and large radius, whereas detected non-SF emission dominates centrally.

This says more than “environment suppresses SF,” while avoiding the unsupported claim that every galaxy follows a single monotonic slope.

11. What is missing from the combined notebook

The 2026-09-04 notebook successfully supplies common stage curves, individual curves, S/N-selected maps, and exact category accounting. It still lacks several pieces needed for a paper-level inference:

1. **The plotted post-cut numerical tables are not displayed.** Cell 23 displays only the uncut tables even though cells 19 and 21 plot the cut tables.
 2. **No uncertainty bands or galaxy bootstrap are included.** Thousands of spaxels cannot substitute for the number of independent galaxies.
 3. **No bin-by-bin support curve is shown.** A text range such as “galaxies/bin: 1--5” hides which bins are supported by one galaxy.
 4. **No explicit common-support mask is enforced in the notebook figures.** Radii beyond roughly $2-2.5R_e$ rapidly become stage- and galaxy-specific.
 5. **No shape statistic is computed.** A truncation/edge measure, inner--outer contrast, or flexible profile model is preferable to one straight slope.
 6. **No leave-one-galaxy-out analysis is present,** including the explicitly requested NGC4383 exclusion.
 7. **The S/N-excluded area disappears from the curve denominator.** The maps show gray, but the curves need either a fourth category or paired cut/uncut presentation.
 8. **No individual-concordance summary is given.** The 26 figures exist, but the notebook does not count how many galaxies share the stage-level sign or edge change.
 9. **Radius and Σ_* are not separated.** A two-dimensional or matched analysis is required.
 10. **The analysis is cross-sectional.** Stage differences assume the classified samples represent an evolutionary ordering; they are not direct temporal transitions.
 11. **No control for mass, morphology, spatial resolution, cluster history, or coverage is included.** These are explicitly acknowledged in the notebook prose but not modelled.
 12. **NSF remains physically unresolved.** The notebook has useful first-failed-gate maps, but no population-level separation of central LI(N)ER/old-star/AGN-like emission from outer shocks/DIG/outflow/tail candidates.
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12. Recommended next analysis for the occupancy question

Before moving to surviving-SF intensity, the minimal strong version of the current fraction analysis should be:

1. Use **equal-galaxy profiles as primary** and pooled-spaxel profiles as a clearly labelled sampling comparison.

2. Show **uncut and S/N-cut results together**. If the cut is retained, add low-S/N excluded as a fourth state relative to the original valid denominator.
3. Plot $N_{\text{gal}}(b)$ directly below each curve and define a preregistered common-support rule, for example at least three galaxies per stage, while testing stricter choices.
4. Add **whole-galaxy bootstrap bands** and stage-difference curves $\Delta f_c(R)$ and $\Delta f_c(\Sigma_\star)$.
5. Replace the one-slope question with a small set of interpretable per-galaxy metrics: inner normalization, outer-minus-inner contrast, and an SF-edge/half-occupancy radius with uncertainty.
6. Run **leave-one-galaxy-out curves for every system**, with NGC4383 reported explicitly. Also test an NGC4383-specific outflow mask rather than assuming that continuum S/N is equivalent to an outflow mask.
7. Fit a galaxy-level or hierarchical binomial/multinomial model with stage-by-radius and stage-by- $\Sigma_\star$ terms, keeping galaxy as the independent cluster. A matched two-dimensional grid is an acceptable descriptive alternative.
8. Split NSF by spatial/diagnostic context: central versus outer, BPT class, EW, intrinsic dispersion, and recognisable outflow/tail/ring/bar morphology.

These steps would turn the current visual result into a defensible statement about whether the stage evolution is edge contraction, normalization loss, or both.

13. Deferred branch: intensity of the surviving SF regions

The fraction analysis cannot answer whether the remaining SF regions are enhanced or suppressed. That is a separate conditional quantity:

$$\Delta \log \Sigma_{\text{SFR}} = \log \Sigma_{\text{SFR}} - \log \Sigma_{\text{SFR, pre}}(\Sigma_\star),$$

conditional on SF.

The later analysis should compare this residual at fixed $\Sigma_\star$ and radius using a frozen pre-peak reference, equal-galaxy weighting, and whole-galaxy resampling. It should distinguish four possibilities:

SF occupancy	Surviving-SF intensity	Interpretation
lower	unchanged	regions disappear; survivors remain normal
lower	lower	fewer regions and weaker survivors
lower	higher	truncation plus compression/enhancement
unchanged	lower	widespread fading without much class loss

Nothing in the present fractions alone selects among these intensity outcomes.

14. Final conclusions

1. **Strong endpoint result:** pre-peak to post-peak is a broad individual-galaxy shift from SF occupancy to ND, not a pooled-spaxel artefact.
 2. **Spatial result:** the SF footprint contracts, with outer/low- $\Sigma_{\star}$ regions preferentially becoming ND and central regions increasingly occupying NSF.
 3. **Geometry:** the best description is **outside-in truncation plus a later disc-wide occupancy decrease**, not pure uniform or pure one-slope outside-in quenching.
 4. **Transition:** close-to-peak is heterogeneous and overlaps both endpoint populations; it should not be treated as a narrow deterministic middle state.
 5. **Individual galaxies:** global endpoint ordering is strong, but radial shapes are diverse because of truncation strength, floor effects, rings, asymmetries, bars, and outflows.
 6. **NSF:** no secure monotonic global stage trend is established. Its central/outer redistribution is potentially informative but requires diagnostic decomposition.
 7. **S/N selection:** it removes unreliable faint regions, especially in NGC4383, but preferentially censors ND and changes the denominator. It is a sensitivity test, not a replacement for an explicit NGC4383/outflow exclusion.
 8. **Next step:** complete the occupancy robustness tests above; then proceed to the independent surviving-SF intensity analysis.
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Appendix A. Per-galaxy values after the S/N cut

Retained is the fraction of the original valid-disc footprint satisfying $S/N_{\text{postfit}} \geq 25$. The final column is $f_{\text{SF}}(1 \leq R/R_e < 2) - f_{\text{SF}}(R < R_e)$.

Galaxy	Stage	Retained	f_{SF}	f_{ND}	f_{NSF}	Outer - inner f_{SF}
NGC4189	pre-peak	0.682	0.667	0.052	0.281	+0.222
NGC4254	pre-peak	0.611	0.731	0.013	0.256	-0.042
NGC4294	pre-peak	0.688	0.547	0.037	0.417	-0.156
NGC4321	pre-peak	0.814	0.475	0.107	0.417	+0.201
NGC4351	pre-peak	0.435	0.380	0.407	0.213	-0.430
NGC4383	pre-peak	0.227	0.282	0.000	0.718	-0.257
NGC4535	pre-peak	0.903	0.352	0.233	0.415	+0.323
NGC4567_8	pre-peak	0.838	0.515	0.250	0.235	-0.344
NGC4298	close-to-peak	0.467	0.595	0.220	0.185	-0.211
NGC4424	close-to-peak	0.838	0.062	0.676	0.262	-0.191
NGC4501	close-to-peak	0.979	0.387	0.211	0.403	+0.019
NGC4654	close-to-peak	0.808	0.602	0.177	0.221	-0.269
NGC4694	close-to-peak	0.879	0.013	0.649	0.339	-0.052
IC3392	post-peak	0.603	0.244	0.597	0.160	-0.530
NGC4064	post-peak	0.714	0.025	0.710	0.264	-0.038
NGC4293	post-peak	0.970	0.002	0.748	0.250	-0.003
NGC4380	post-peak	0.465	0.240	0.541	0.219	-0.067
NGC4394	post-peak	0.967	0.174	0.567	0.259	+0.149
NGC4405	post-peak	0.441	0.172	0.683	0.145	-0.711
NGC4419	post-peak	0.741	0.043	0.484	0.473	-0.017
NGC4450	post-peak	0.941	0.036	0.690	0.273	+0.017
NGC4457	post-peak	0.743	0.022	0.518	0.460	+0.006
NGC4569	post-peak	0.971	0.071	0.677	0.252	-0.169
NGC4580	post-peak	0.547	0.227	0.572	0.201	-0.474
NGC4606	post-peak	0.846	0.022	0.867	0.111	-0.046
NGC4698	post-peak	0.635	0.029	0.507	0.464	+0.038

Appendix B. Reproducibility and scope

Derived numeric tables and report figures are stored in `20260905_stage_SF_fraction_analysis_assets/`. The calculation-only audit executed the combined notebook's configuration, helper, inventory, counting, estimator, and S/N-estimator cells against the live products. It did not execute the 52 per-galaxy plotting loops again and did not change any source notebook. Stored notebook PNGs were extracted byte-for-byte for the spatial atlases and NGC4383 figures.

The additional bootstrap, annular contrasts, pairwise ordering, edge metrics, and leave-one-out values are report-level diagnostics, not claims that the source notebook already contained those tests.